Charlton Smith

Principal Software Engineer & AI Consultant

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Independent consultant focused on AI applications and production software. Experience spans enterprise engineering, startup product development, and the design and operation of independent AI and full-stack products.

ENGINEERING EXPERIENCE

Principal Software Engineer — Current engagement

Independent consulting

Working with a client in a principal software engineering role with a strong focus on AI.

Independent Software Engineer — Nov 2023 – Present

Self-employed

- Build and operate AI and full-stack products with shared retrieval infrastructure, APIs, and activity logging.
- Built EmbedRoute, an embedding gateway connecting four providers through one API; operate agent workflows with per-call cost and error logging.

Senior Software Engineer — Nov 2022 – Nov 2023

Capital One

- Developed Python microservices for test automation and enhanced testing pipelines using AWS Lambda and S3.
- Mentored an intern who shipped a test automation tool.

Software Engineer II — Jun 2018 – Apr 2022

T-Mobile

- Built internal automation that reduced manual workload by approximately 80%.
- Designed Python/Django backend services and migrated a frontend from Django templates to Angular.

Web3 Automation Engineer — Sep 2021 – Jun 2022

Dustbunny · Independent project

Built a distributed bidding system with Dockerized workers and Redis coordination, including polling, caching, and operational controls.

Co-Founder & Technical Lead — Jan 2016 – Mar 2018

OLI Fitness

Built Kinect-based computer vision for exercise feedback. Published an ACM CHI 2017 extended abstract and reached the Princeton Tiger Launch finals.

Charlton Smith

Selected work & capabilities

SELECTED INDEPENDENT WORK

Products and infrastructure I have designed, built, and operated.

EmbedRoute

azoni.ai/projects/embedroute

An OpenAI-compatible embedding gateway with provider routing, request caching, API-key scoping, and per-call cost logging. A shared API for four providers, used by the azoni.ai assistant and Old Ways Today.

AI infrastructure · TypeScript · Next.js · Supabase · PostgreSQL

FaB Stats

azoni.ai/projects/fab-stats

A community analytics platform for competitive card-game players. Connected tournament imports, match analytics, cached public summaries, and a Discord companion to reduce the friction of collecting and using match data.

Product & data systems · Next.js · TypeScript · Firestore · Netlify Functions

Old Ways Today

azoni.ai/projects/old-ways-today

A content and recommendation product with an assistant grounded in a curated knowledge base. Integrated retrieval, streaming responses, and product-link validation with fallback handling.

Applied AI · Next.js · FastAPI · Firestore · EmbedRoute

Shared agent infrastructure

azoni.ai/projects/azoni-mcp

Centralized agent tools behind authenticated, rate-limited endpoints with common activity logging. Supports chat and autonomous workflows through a shared server organized by domain.

Agent systems · Node.js · Express · Firebase Admin · Render

TECHNICAL FOCUS

AI applications	Retrieval, embeddings, AI assistants, agent workflows, tool integrations, and cost visibility
Product engineering	Python, TypeScript/JavaScript, React, Next.js, Node.js, FastAPI, Django, and REST APIs
Data & infrastructure	PostgreSQL, Firebase, Redis, AWS Lambda/S3/EC2, Docker, and CI/CD

EDUCATION

M.S. Software Engineering

Colorado Technical University · 2021

Completed while working full-time

B.S. Computer Science

University of Washington Tacoma · 2017

Graduated with Honors